

The photo is the proof of work. Today, nothing proves the photo.

Context · allegation

Kandhamal, Odisha · reported 1 July 2026

Odisha Vigilance arrested two MGNREGS engineers over an **alleged** fraud of more than ₹42 lakh. According to OmmCom News, “mandatory digital geo-tagging of the project sites at various stages ... was never done.”

Arrests on allegation, not convictions (OmmCom News; OrissaPOST). PRAMAAN cannot force a capture to happen.

Core evidence

CAG Performance Audit of MGNREGS, Karnataka · Report No. 13 of 2025

The CAG found geo-tagged photos “captured from existing photograph”, a photo of a different shed, and the same photos reused across work stages, used to release payments.

CAG test-check findings from sampled gram panchayats, not state-wide totals. Not PRAMAAN results.

PRAMAAN is designed to make a field capture independently verifiable as evidence at the moment it is captured: a phone app that checks a 3-second clip on the device and issues a Certificate that a second party can check on a laptop Verifier Desk.

Everything shown in this deck is a presentation-only prototype or a proposed implementation for the event. Nothing has been built, run or measured yet.

PROBLEM

In MGNREGS the site photo releases payment, and auditors have documented it being faked in simple ways

Schemes such as PMAY and MGNREGA rely on geo-tagged photographs as proof of physical work. EXIF metadata and manual review are weak against low-skill attacks.

CAG FOUND CASES WHERE... (REPORT NO. 13 OF 2025)	PATTERN	PRAMAAN DESIGN ELEMENT IT IS AIMED AT (PROPOSED)
"photograph captured from existing photograph was uploaded" (p. 116)	Photo of a photo (recapture)	FFT-based Moiré / Recapture Detection, usually reinforced by Motion Consistency
"photograph of another existing waste management shed was uploaded" (pp. 47–48)	Photo of a different site	Geofence / Location Binding
"same photographs were uploaded for different stages of work" (pp. 45–46)	Same photos reused across stages	Camera-only capture with no gallery upload. Not a detection check

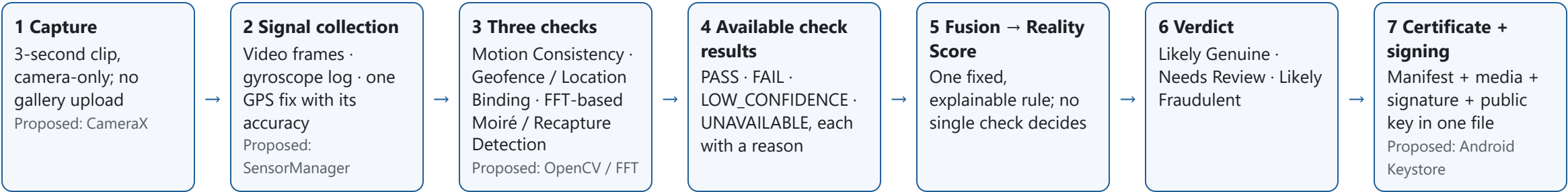
CAG test-check findings from sampled gram panchayats, not state-wide totals, and not results from PRAMAAN.

Check the capture on the phone, at the moment it is taken, and seal the result in a Certificate

Proposed implementation for the event

 Nothing below is built yet. Technology names are labels only.

Phone app · on-device and offline



Hand-over: one Certificate file, phone to laptop, no cloud service; transfer method chosen in the event build.

8 Verifier Desk (laptop): designed to open the Certificate, check the signature, and show the verdict, Reality Score, each check's reason and the evidence.

Deterministic signal processing, no trained model. App-level signing, not hardware-attested.

STAGE	IN THIS PROTOTYPE	PROPOSED FOR THE EVENT BUILD
Capture, signals, checks	Screens with a camera placeholder; check names only; no camera, GPS, sensor or FFT processing	3-second clip, gyroscope and GPS on the loaner iQOO 15; the three checks written during the event
Fusion, verdict	Hardcoded sample results, chosen by the reviewer	Fixed-rule fusion, tuned during the event
Certificate, Verifier Desk	Illustrative Certificate; no key, hash or signature; concept screen, nothing verified	App-level signing with Android Keystore; laptop app that checks the signature and shows the evidence

No single check decides. Uncertainty is not fraud.

Motion Consistency is designed to ask whether the phone's own movement, from the gyroscope, matches the movement seen in the clip.

Geofence / Location Binding is designed to measure how far the capture is from the claimed registered site.

FFT-based Moiré / Recapture Detection is designed to look for the regular patterns that appear when a screen is filmed.

Each check will report **PASS**, **FAIL**, **LOW_CONFIDENCE** or **UNAVAILABLE**. An **UNAVAILABLE** check is left out; **LOW_CONFIDENCE** is uncertainty, not a sign of fraud.

Prototype interaction — simulated result

The reviewer picks each sample result; nothing is computed.

Prototype interaction — simulated result

✓ Likely Genuine

This capture shows the characteristics of a genuine on-site capture.

Reality Score 88 sample value, prototype interaction

Based on all three checks.

Motion Consistency

PASS

Phone movement matches the movement seen in the clip.

Geofence / Location Binding

PASS

Captured 6.4 m from the claimed registered site.
sample value, prototype interaction

FFT-based Moiré / Recapture Detection

PASS

No screen-pattern signs in the clip.

PRAMAAN proves capture authenticity, not scene truthfulness.

View certificate

SC-07 · Likely Genuine sample

Prototype interaction — simulated result

⚠ Needs Review

This capture routes to human follow-up.

Reality Score 61 sample value, prototype interaction

Based on the two available checks; one check was UNAVAILABLE.

Motion Consistency

UNAVAILABLE

This check could not run for this capture. It is not counted as a failure; the result uses the remaining checks.

Geofence / Location Binding

LOW_CONFIDENCE

GPS accuracy was 30 m, too coarse to confirm the site. This is uncertainty, not a sign of fraud.
sample value, prototype interaction

FFT-based Moiré / Recapture Detection

PASS

No screen-pattern signs in the clip.

PRAMAAN proves capture authenticity, not scene truthfulness.

View certificate

SC-07 · Needs Review sample

Prototype interaction — simulated result

✗ Likely Fraudulent

This capture could not be confirmed as a genuine on-site capture.

Reality Score 24 sample value, prototype interaction

Based on all three checks.

Motion Consistency

FAIL

Phone movement does not match the movement seen in the clip.

Geofence / Location Binding

PASS

Captured 5.1 m from the claimed registered site.
sample value, prototype interaction

FFT-based Moiré / Recapture Detection

FAIL

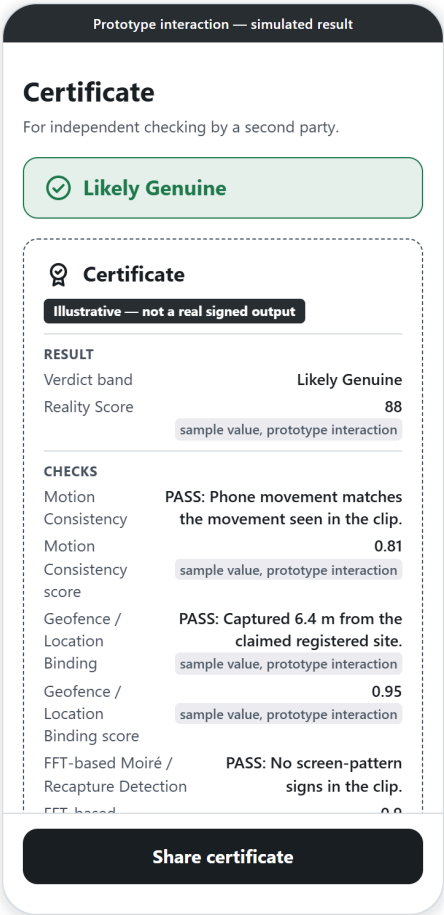
Screen-pattern signs in the clip, as when a screen is filmed.

PRAMAAN proves capture authenticity, not scene truthfulness.

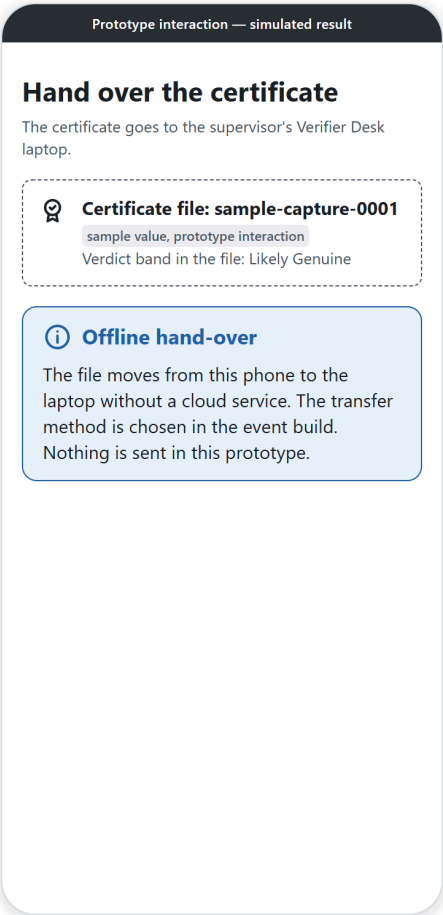
View certificate

SC-07 · Likely Fraudulent sample

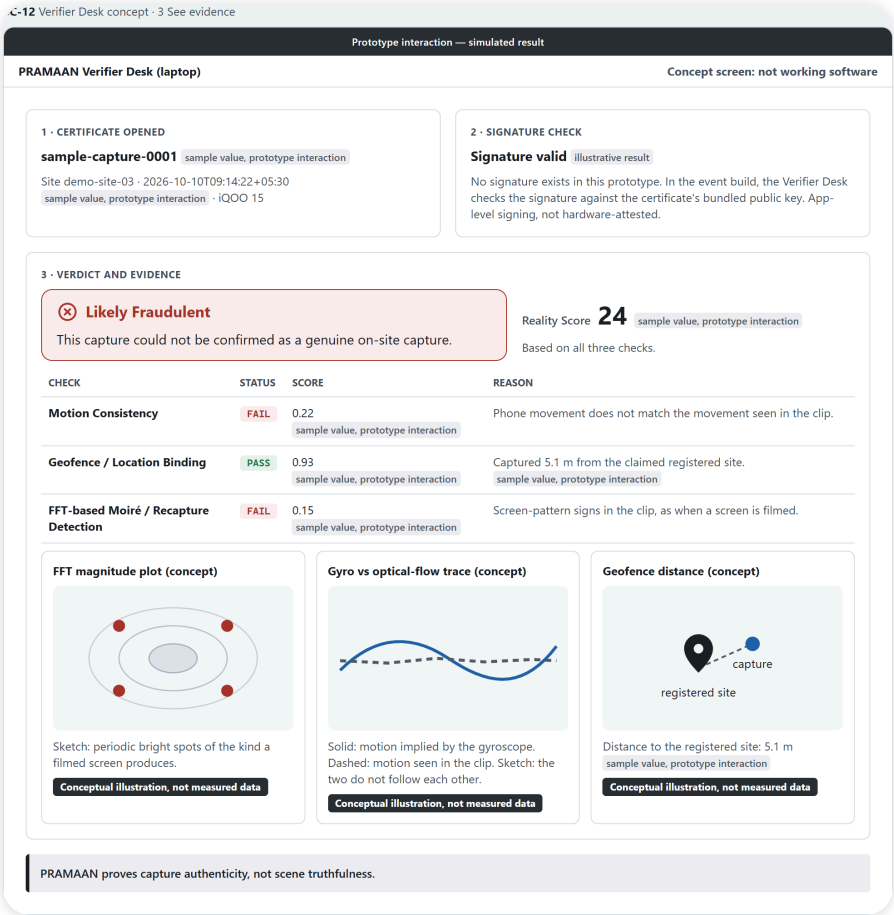
A Certificate a second party can check on a laptop



SC-08 Certificate · Illustrative — not a real signed output



SC-09 Offline hand-over · nothing is sent



SC-12 Verifier Desk concept · evidence views are conceptual illustrations, not measured data

Prototype interaction **Conceptual illustration** In the prototype no key, hash or signature exists; the signature result is illustrative. **Proposed for the event** App-level signing with Android Keystore, not hardware-attested; a laptop Verifier Desk that checks the signature and shows the evidence.

Location trust tells you where the phone is.

PRAMAAN is designed to tell you whether the capture is genuine.

	ANCHOR (FROM ITS PUBLIC ONE-LINER)	PRAMAAN (PROPOSED)
Question answered	Can this phone's location be trusted?	Was this capture genuinely made live, here, now, by this camera?
Object of trust	The device's position	The captured media
Location's role	The whole answer	One of three checks (with Motion Consistency and FFT-based Moiré / Recapture Detection), fused together
Output	Not publicly stated	A Certificate a second party can check on the Verifier Desk

Anchor, Team Nexus, 1st Runner-Up (Students), Bengaluru City Battle: it “verifies whether your phone's location can be trusted by cross-checking GNSS with physical sensors and on-device AI, even without internet connectivity” (iQOO Community, thread 169162). Both are offline and on-device; Geofence / Location Binding is where we overlap most. Layers, not rivals: location trust is necessary, not sufficient.

Not known and not claimed: Anchor's model, sensors, accuracy or UI, or whether it inspects media or issues certificates.

PRAMAAN proves capture authenticity, not scene truthfulness.

It cannot tell whether a structure built at the correct site meets the scheme's criteria.

Also stated openly

- GPS spoofing with a mock-location app is a known gap; v2 mitigation: Wi-Fi / cell-tower cross-check.
 - Reuse of an earlier genuine capture of the same site is outside what a single capture check can see; camera-only capture raises the bar but does not detect it.
 - App-level signing, not hardware-attested: a rooted or modified device can bypass it; hardware attestation is roadmap only.
- FFT-based Moiré / Recapture Detection will be tuned against specific screens at the venue, listed, with no claim of general validity.
 - Indoors, GPS can be slow or imprecise; poor GPS or a textureless scene is designed to give LOW_CONFIDENCE.
 - The event build will use a few hardcoded demo sites, not a site-registration system. PRAMAAN cannot force a capture to happen.

Every capability below will be built during the event, on the iQOO 15 and a laptop

Red Light · phone only (PRD §11.1)

0–2	Demo-site registry, camera skeleton
2–6	Geofence / Location Binding (built first)
6–14	Motion Consistency
14–18	FFT-based Moiré / Recapture Detection
18–22	Fusion → Reality Score; Certificate with app-level signing
22–26	Integration; per-check fault isolation (UNAVAILABLE without a crash)

Green Light · phone + laptop (PRD §11.2)

26–30	Tune FFT-based Moiré / Recapture Detection on venue screens; list the screens
30–36	Verifier Desk on the laptop
36–40	Full rehearsal: screen-recapture and wrong-location cases
40–48	Buffer, final rehearsal; numbers recorded at the event, not before

Stretch (PRD §10, no hours)

Optional on-device VLM, only after the three-check core runs unattended twice, never feeding the Reality Score. Torch / light challenge, tested on venue lighting first. Hardware attestation: v2 only.

WHAT MAKES US STAND OUT · TEAM

Offline, on-device trust tools for location already exist; at the Bengaluru City Battle, Team Nexus's Anchor was one. PRAMAAN targets a different object: the capture itself. In MGNREGS the geo-tagged photograph is the proof that releases public money, and the CAG's 2025 Karnataka audit found photos “captured from existing photograph”, a photo of a different shed, and the same photos reused across stages. A correct location alone does not show whether an image was captured live or recaptured from a screen. PRAMAAN is designed to fuse three checks, Motion Consistency, Geofence / Location Binding and FFT-based Moiré / Recapture Detection, into a certificate a second party can check independently. The checks, the certificate signing and the Verifier Desk will all be built live during the event, the phone app on the iQOO 15.

Built before the event (disclosed)

Product concept and PRD; UX journey; a presentation-only clickable web prototype with no detection, scoring, camera, sensor, location, signing or verification logic; proposed architecture; conceptual visuals; illustrative Certificate; evidence research.

Built during the event

The Android app on the loaner iQOO 15, the three checks, fusion and Reality Score, Certificate signing, the Verifier Desk, and all testing and measurement.

Team · prior builds & hackathons

[TEAM INPUT PENDING] Not drafted; must come from each team member.

PRAMAAN proves capture authenticity, not scene truthfulness.